

Examination Information Page

Written examination

General information about the exam:

Subject code: FM1015

Subject name: Modelling of Dynamic Systems

Examination date: 01.12.2023

Examination time: 09:00-12:00

Total hours: 3

Responsible course manager: Bernt Lie

Campus: Porsgrunn

Faculty: Technology, Natural Sciences and Maritime Sciences

No. of assignments: 20 Problems (= questions)

No. of attachments: 1

No. of pages incl. information page and attachments: front page + 16 (1-17)

Permitted aids:

No aids permitted. [Except PC to get access to this document in FLOWhandin + Excel file where answers must be put, and use of FLOWhandin to upload answers.]

Information regarding attachments:

Multiple Choice exam: insert answers (A-D) in attached Excel sheet, and upload in Excel file format.

Comments:

Your answers to Problem 1-20 (A-D) must be inserted into Excel sheet, and uploaded as .xlsx file.

Exam assignment:

Each of 20 Problems has optional answers A, B, C, D. Make your choice of A-D.

If the exam isn't in digital form, please select the type of examination paper:

Spreadsheets ☐

Line sheets ☐

THE CANDIDATE MUST VERIFY THAT THE TASK SET IS COMPLETE ON THEIR OWN.

Exam: FM1015 Modelling of Dynamic Systems

Date: Friday, December 1, 2023

Form of exam: This is a *Digital Exam* of type *Multiple Choice*. Each exam *question* (denoted “problems” in the text below) is followed by *4 possible answers* (“choice” in the tables related to questions). *Some* details about your answers and how they will be assessed:

- A single correct choice gives full score on a problem; any other choice (an incorrect choice, no choice, several choices) gives zero score on the particular problem.
- All problems are given the same weight.
- Most problems are posed so that it is necessary to do background calculations to choose the correct answer (A–D).
- The background calculations shall not be submitted — the only submission in the Digital Exam are the choices (A–D) in each problem.
- You need to insert the answers in the *attached Excel* sheet, and upload this *Excel* sheet which includes your answers, in the exam system.
- The file you **upload must be in Excel file format**, i.e., *.xlsx format. (*Reason:* the files will be automatically parsed for grading.)
- A total score of 0-39% gives grade F (fail), score 40-49% gives grade E, score 50-59% gives grade D, score 60-79% gives grade C, score 80-89% gives grade B, and score 90-100% gives grade A.

Aids: No aids allowed, except the necessary use of PC for using WISEflow, and the use of Excel for filling in answers.

This exam contains questions (denoted *problems*) in the *Problems* part. The Problems part consists of *20 problems*, each with 4 alternative choices (A–D).

Contact person for questions during the exam:	
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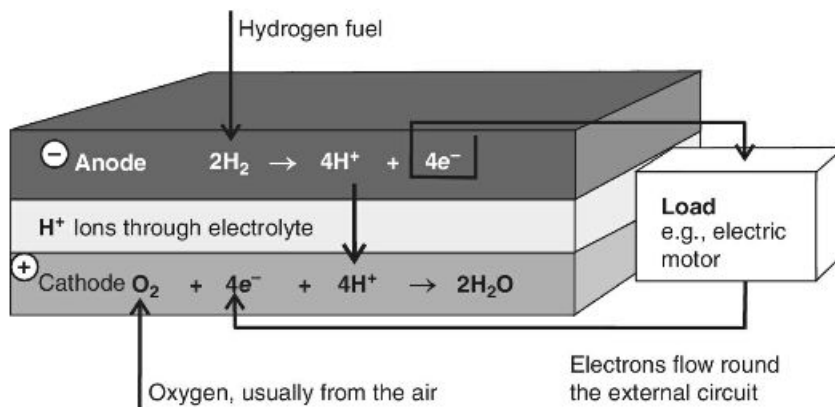


Figure 1: Fuel Cell with acid electrolyte and hydrogen fuel.

Problems

Hydrogen (H_2) and oxygen (O_2), when mixed together, form an explosive mixture¹ which reacts quickly and violently. If mixed with other, inert gases such as nitrogen (e.g., in air), the mixture is less explosive, and may become harmless when the partial pressure of hydrogen is below a certain threshold value.

It is of interest to try to control this violent reaction so that we can utilize the energy in the mixture. This can be done by putting the reacting gases (H_2 , O_2) in different volumes, separated by a *membrane*. Such a membrane may be physically damaged if the *total pressure difference across* the membrane is too large (for the fuel cell discussed below, the pressure difference should be less than ca. 0.2 bar). Also, if the *partial pressure difference* for one gas is large across the membrane, this particular gas may “leak” (permeate) through the membrane.

A fuel cell can be used to take out the energy of hydrogen and oxygen: a certain type of membrane that under normal conditions blocks permeation of any species except *proton* (H^+) can be used (PEM = Proton Exchange Membrane). Then some half-reactions occur on each side of the membrane, accelerated by a catalyst. The fuel cell includes a system for feeding of reactants (H_2 , O_2) to each side of the membrane, and for removing the product (H_2O). Note that normal PEM systems require the membrane to be humid (= contain some water) in order to work. Because of this, some water is normally added to both sides of the membrane.

Problem 1. Reactions in Fuel Cell.

Consider a Fuel Cell where hydrogen fuel (H_2) is added to the so-called *anode volume*, while oxygen (O_2) is added to the *cathode volume* (typically via air), where an (acid electrolyte) membrane separates the two volumes and only allows for transport of *proton* (H^+) from the anode side to the cathode side, Fig. 1.

At the *anode* interface between gas, some catalyst, and the proton conducting membrane (the so-called triple-point junction), hydrogen H_2 reacts in a *half-cell reaction* as follows:



¹A $\text{H}_2 + \text{O}_2$ mixture is sometimes known as “knallgas” — taken from German and Scandinavian languages = bang-gas

Proton H^+ can move through the membrane towards the *cathode* interface. The *electron* e^- , on the other hand, moves through the external electric circuit and eventually ends up at the cathode.

At the *cathode* triple-point junction, oxygen O_2 reacts with the proton coming through the electrolyte and the electron coming via the external electric circuit in the following *half-cell reaction*:



Combining the two half-cell reactions, the overall reaction is sometimes written as



where the product is *water*, but other possibilities exist.

Let the species be ordered according to the vector $(\text{H}_2, \text{O}_2, \text{H}_2\text{O})$. It is of interest to formulate the *stoichiometric matrix* ν for the overall reaction in Eq. 3, as well as the *positive stoichiometric parameter* for the *electron*, ν_{e^-} (Note: $\nu_{\text{e}^-} > 0$).

	Problem 1: Which of the following pairs of ν and ν_{e^-} is <i>incorrect</i> ?	Choice
A:	$\nu = (1, -\frac{1}{2}, 1)$ and $\nu_{\text{e}^-} = 2$	
B:	$\nu = (-1, -\frac{1}{2}, 1)$ and $\nu_{\text{e}^-} = 2$	
C:	$\nu = (-\frac{1}{2}, -\frac{1}{4}, \frac{1}{2})$ and $\nu_{\text{e}^-} = 1$	
D:	$\nu = (-2, -1, 2)$ and $\nu_{\text{e}^-} = 4$	

▲

Problem 2. Equilibrium voltage in Fuel Cell

The so-called *chemical potential* μ_j of species j in a mixture is given as molar Gibbs' energy $\tilde{G}_j^\circ$ of species j at standard state plus a term $RT \cdot \ln a_j$ where R is the gas constant, T is absolute temperature (i.e., in K), and a_j is the *activity*. With $\mu_j \equiv \tilde{G}_j$, we have

$$\tilde{G}_j = \tilde{G}_j^\circ + RT \cdot \ln a_j. \quad (4)$$

For *ideal gas*,

$$a_j = \frac{p_j}{p^\circ} \quad (5)$$

where p_j is the partial pressure of species j , while p° is a standard state *scaling* pressure (normally either 1 atm or 1 bar). In the liquid case, for an *ideal solution*

$$a_j = x_j \quad (6)$$

where x_j is the mole fraction $x_j = n_j/n$ of the liquid.

Next, the molar Gibbs energy of *reaction* $\Delta_r \tilde{G}$ (at arbitrary T , p_j) is given as

$$\Delta_r \tilde{G} = \sum_j \nu_j \tilde{G}_j, \quad (7)$$

while the *standard* molar Gibbs energy of reaction (i.e., at arbitrary T and standard pressure p°) is denoted $\Delta_r \tilde{G}^\circ$. The equilibrium voltage from the cathode side (positive) to the anode side (negative) is then

$$u_{\text{n}} = -\frac{\Delta_r \tilde{G}}{F \nu_{\text{e}^-}} \quad (8)$$

where F is *Faraday's number*.

	Problem 2: Equilibrium voltage — which expression is <i>incorrect</i> ?	Choice
A:	$u_n = \left(-\Delta_r \tilde{G}^\circ - RT \sum_j \ln a_j^{\nu_j} \right) / (F \nu_{e^-})$	
B:	$u_n = \left(-\Delta_r \tilde{G}^\circ - RT \ln \left(\prod_j a_j^{\nu_j} \right) \right) / (F \nu_{e^-})$	
C:	$u_n = \left(-\Delta_r \tilde{G}^\circ - RT \prod_j \ln a_j^{\nu_j} \right) / (F \nu_{e^-})$	
D:	$u_n = \left(-\Delta_r \tilde{G}^\circ / \nu_{e^-} - RT \ln \left(\prod_j a_j^{\nu_j / \nu_{e^-}} \right) \right) / F$	

▲

Problem 3. Gas vs. liquid form of water product.

In the group project, it was assumed that water was present in the cathode volume at *saturation pressure*, which is a pure function of temperature:

$$p_{\text{H}_2\text{O}}^c = p_{\text{H}_2\text{O}}^{\text{sat}}(T).$$

Saturation pressure for water may be given by the Antoine equation,

$$\log_{10} \left(\frac{p_{\text{H}_2\text{O}}^{\text{sat}}}{p^\circ} \right) = A_{\text{H}_2\text{O}} - \frac{B_{\text{H}_2\text{O}}}{T + C_{\text{H}_2\text{O}} - 273.15 \text{ K}} \quad (9)$$

where T is absolute temperature (K) and

$$\begin{aligned} A_{\text{H}_2\text{O}} &= 5.11564 \\ B_{\text{H}_2\text{O}} &= 1687.537 \text{ K} \\ C_{\text{H}_2\text{O}} &= 230.17 \text{ K}. \end{aligned}$$

This implies that in the expression for $\tilde{G}_{\text{H}_2\text{O}}$ in Eq. 4, the water activity was set to

$$a_{\text{H}_2\text{O}}^c = \frac{p_{\text{H}_2\text{O}}^{\text{sat}}}{p^\circ}.$$

In much of the relevant published literature, *instead* it is assumed that water is present in the cathode volume as *condensed water*, i.e., water in *liquid form*. Because the other involved species (oxygen, nitrogen, etc.) are (almost) completely in gas form, this implies that the liquid phase is pure water, i.e., $x_{\text{H}_2\text{O}}^c \approx 1$, and the activity of water is unity:

$$a_{\text{H}_2\text{O}}^c = x_{\text{H}_2\text{O}}^c = 1.$$

An interesting question is: *can both assumptions on cathode water state* lead to the same equilibrium voltage, or is the assumption of liquid water leading to $a_{\text{H}_2\text{O}}^c = 1$ just introduced for convenience?

To help us assess this, under the assumption of *constant heat capacity* $\tilde{c}_{p,j}$, we have:

$$\tilde{G}_j^\circ = \Delta_f \tilde{G}_j^\circ - \Delta_f \tilde{S}_j^\circ (T - T^\circ) + \tilde{c}_{p,j} \left(T - T^\circ - T \cdot \ln \frac{T}{T^\circ} \right). \quad (10)$$

For water in *gas phase*:

$$\begin{aligned} \Delta_f \tilde{G}_{\text{H}_2\text{O},g} &= -228.572 \text{ kJ/mol} \\ \Delta_f \tilde{S}_{\text{H}_2\text{O},g} &= 188.7 \text{ J/mol K} \\ \tilde{c}_{p,\text{H}_2\text{O},g} &= 33.58 \text{ J/mol K} \end{aligned}$$

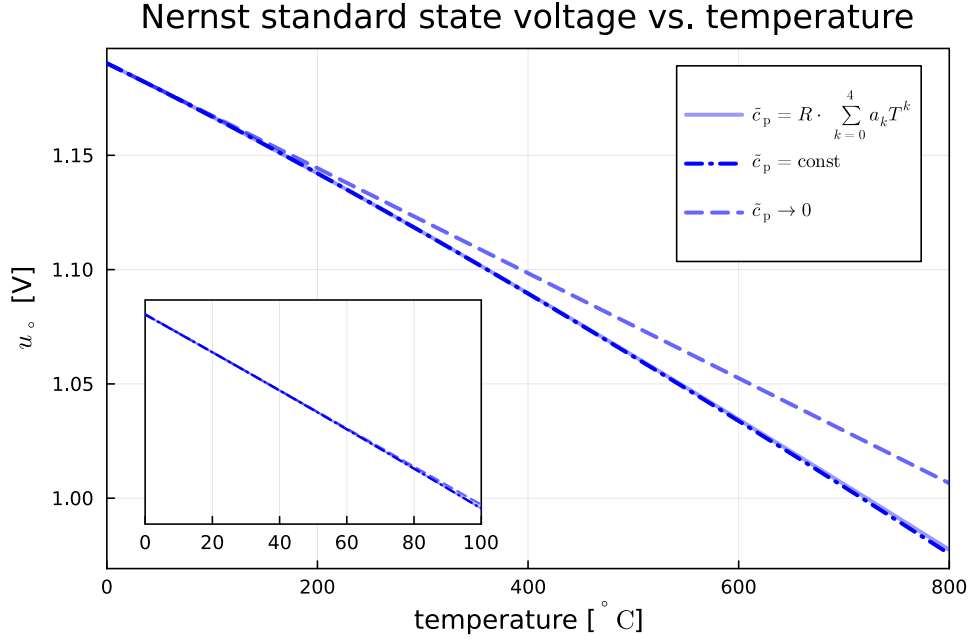


Figure 2: Hydrogen Fuel Cell standard state equilibrium voltage u_o .

while for water in *liquid phase*:

$$\Delta_f \tilde{G}_{\text{H}_2\text{O},\ell} = -237.129 \text{ kJ/mol}$$

$$\Delta_f \tilde{S}_{\text{H}_2\text{O},\ell} = 69.91 \text{ J/mol K}$$

$$\tilde{c}_{p,\text{H}_2\text{O},\ell} = 75.29 \text{ J/mol K}.$$

	Problem 3: Assuming H_2O product in gas or liquid form — which of the following statements is <i>correct</i> ?	Choice
A:	The assumption of H_2O in liquid form is only introduced for <i>convenience</i> , and leads to a slightly different equilibrium voltage u_n compared to assuming gas.	
B:	Whether cathode H_2O is in liquid or gas form, in both cases we get identical equilibrium voltage u_n .	
C:	The assumption of H_2O in gas or liquid form leads to different equilibrium voltages u_n . However, if we take into account the energy balance and <i>condensation</i> of the H_2O in gas form, this will modify the temperature and will gradually lead to the same equilibrium voltage u_n .	
D:	The H_2O content on the <i>anode</i> side will also affect the equilibrium voltage u_n , and needs to be taken into account to get a correct equilibrium voltage u_n .	

▲

Problem 4. Pressure effect on equilibrium voltage.

Assuming that water is produced in *gas phase*, the standard state equilibrium voltage u_o is as in Fig. 2.

If any of the partial pressures p_{H_2} , p_{O_2} , or $p_{\text{H}_2\text{O}}$ deviate from the standard state $p^\circ = 1 \text{ bar}$, there will be a correction term to the equilibrium voltage as indicated in Problem 2.

We will assume that cathode water is at the saturation pressure (water is produced at the cathode), which for $T = 80^\circ\text{C}$ is $p_{\text{H}_2\text{O}}^c \approx p_{\text{H}_2\text{O}}^{\text{sat}} \approx 0.4 \text{ bar}$, while the anode water pressure is $p_{\text{H}_2\text{O}}^a \approx \frac{1}{2}p_{\text{H}_2\text{O}}^{\text{sat}} \approx 0.2 \text{ bar}$ (the membrane must contain some water to work, and some water is normally added to the anode volume).

For PEM fuel cells, the total anode pressure p_a and total cathode pressure p_c should be equal, and not deviate by more than $\Delta p_{ac} = |p_a - p_c| \leq 0.2 \text{ bar}$ to avoid mechanical damage to the membrane. Typical electrode pressures p_a, p_c lie in the range of $[1, 5] \text{ bar}$.

Assume that we operate the PEM fuel cell with electrode pressures $p_a = p_c = 5 \text{ bar}$, that we feed the anode with pure hydrogen (+ some water), while we feed the cathode with air (+ some water) where the oxygen mole fraction is approximately 0.2. Assume that we feed sufficient air to the cathode side to maintain mole fraction oxygen of 0.2 in the cathode volume.

What is the approximate change u_p in equilibrium voltage away from the standard state voltage (u_o ; Fig. 2) due to pressure conditions as indicated above? Use gas constant $R = 8.31 \text{ J}/(\text{mol} \cdot \text{K})$, Faraday's constant $F = 9.648 \cdot 10^4 \text{ C/mol}$, and assume operation at T given by 80°C .

	Problem 4: Approximate change u_p in equilibrium voltage due to pressure conditions specified above is	Choice
A:	$u_p \approx 0.12 \text{ V}$	
B:	$u_p \approx -33 \text{ mV}$	
C:	$u_p \approx 0.31 \text{ V}$	
D:	$u_p \approx 37 \text{ mV}$	

▲

Problem 5. Transfer function for Activation voltage loss.

On the convoluted interface between a charged eletrode and the electrolyte, there will be a concentration of ions in the electrolyte with opposite charge as to that of the electrode. This creates a *capacitance*, and there will be such a capacitance on both electrodes. In general, capacitance is given by

$$C_{dl} = \varepsilon_{dl} \frac{A_{dl}}{\delta_{dl}} \quad (11)$$

where ε_{dl} is the *permittivity* of the space between the ions in the double layer, A_{dl} is the area of the double layer, and δ_{dl} is the separation distance between the ions in the double layer. Due to large area A_{dl} and short distance δ_{dl} , the capacitance C_{dl} can be quite large, and in the order of a few Farads. Theoretical expressions exist for C_{dl} .

In the group project, activation voltage loss was only considered on *one* electrode. Figure 3 shows a linearized electric circuit equivalent of the membrane kinetics, with external current i , activation resistances R_a^a, R_a^c at the anode and cathode electrode, respectively (given by linearizing the Butler-Volmer equation, or similar), double layer capacitors C_{dl}^a, C_{dl}^c at the anode and cathode electrodes, respectively, and membrane resistance R_m . Figure 3 shows the relevant currents i (green color), i_r^a, i_{dl}^a, i_r^c , and i_{dl}^c (red color), and as well as relevant voltages u_a^a, u_r , and u_a^c (blue color). Because the depicted circuit in Fig. 3 is linear, the order of the blocks is irrelevant, and we will neglect the membrane resistor R_m in the sequel; we will set the *total* activation voltage u_a equal to the *sum* of activation voltage losses on the anode (u_a^a) and the cathode (u_a^c). Because the course syllabus for course FM1015 Modelling of Dynamic Systems does not include modeling of electric circuits, the model for the activation voltages is given below for

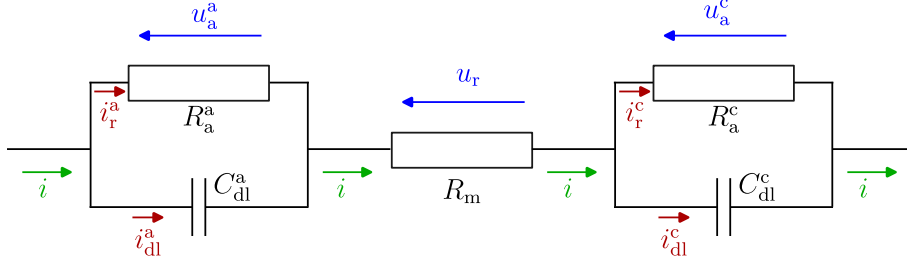


Figure 3: Equivalent linear electric model for kinetics of membrane.

convenience:

$$C_{dl}^a \frac{du_a^a}{dt} = i_{dl}^a \quad (12)$$

$$u_a^a = R_a^a i_r^a \quad (13)$$

$$i = i_r^a + i_{dl}^a \quad (14)$$

$$C_{dl}^c \frac{du_a^c}{dt} = i_{dl}^c \quad (15)$$

$$u_a^c = R_a^c i_r^c \quad (16)$$

$$i = i_r^c + i_{dl}^c \quad (17)$$

$$u_a = u_a^a + u_a^c. \quad (18)$$

It is of interest to find the *transfer function* $h(s)$ from external current i to total activation voltage loss u_a , i.e.,

$$u_a(s) = h(s) \cdot i(s).$$

	Problem 5: Which expression for the transfer function from $i(s)$ to $u_a(s)$, denoted $h(s)$, is <i>correct</i> ?	Choice
A:	$h(s) = \frac{R_a^a}{1+R_a^a C_{dl}^a s} \cdot \frac{R_a^c}{1+R_a^c C_{dl}^c s}$	
B:	$h(s) = (R_a^a + R_a^c) \frac{1+(C_{dl}^a+C_{dl}^c)s}{(1+R_a^a C_{dl}^a s)(1+R_a^c C_{dl}^c s)}$	
C:	$h(s) = (C_{dl}^a + C_{dl}^c) \frac{R_a^a R_a^c + (R_a^a + R_a^c)s}{(1+R_a^a C_{dl}^a s)(1+R_a^c C_{dl}^c s)}$	
D:	$h(s) = \frac{R_a^a}{1+R_a^a C_{dl}^a s} + \frac{R_a^c}{1+R_a^c C_{dl}^c s}$	

▲

Problem 6. Double layer capacitance.

Using a more accurate but more complex model than the Butler-Volmer equation, step changes in external current i leads to the responses in activation voltage losses u_a^a (Wang-HOR) and u_a^c (Wang-ORR) as depicted in Fig. 4. In particular, observe (i) the differences in activation voltage losses at the electrodes, and (ii) the differences in time constants.

The transfer function in Problem 5 has *two* time constants, $T_a = R_a^a C_{dl}^a$ and $T_c = R_a^c C_{dl}^c$ at the anode and cathode, respectively. In the sequel, assume for simplicity that $C_{dl}^a = C_{dl}^c = C_{dl}$.

Given the large difference in gain and time constants for the transfer functions, propose a *simplified* transfer function $h_{\text{simp}}(s)$ for the activation voltage loss, and suggest the

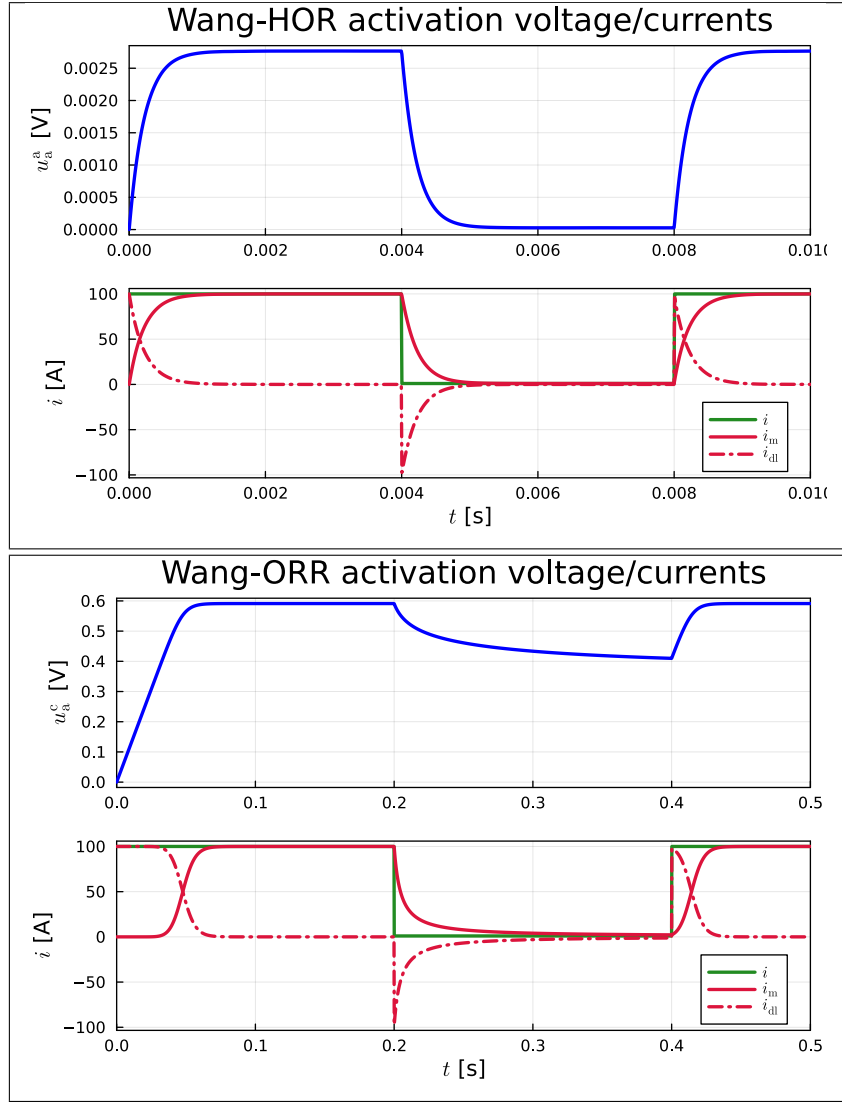


Figure 4: Step responses in activation voltages at the anode (u_a^a) and cathode (u_a^c), respectively.

corresponding simplified capacitor C_{simp} to use.

	Problem 6: Simplified transfer function $h_{\text{simp}}(s)$ with overall capacitor C_{simp} — which of the following pairs is <i>correct</i> ?	Choice
A:	$h_{\text{simp}}(s) = R_a^c / (1 + R_a^c C_{\text{simp}} s)$, $C_{\text{simp}} = C_{\text{dl}}/2$	
B:	$h_{\text{simp}}(s) = R_a^a / (1 + R_a^a C_{\text{simp}} s)$, $C_{\text{simp}} = C_{\text{dl}}/2$	
C:	$h_{\text{simp}}(s) = R_a^c / (1 + R_a^c C_{\text{simp}} s)$, $C_{\text{simp}} = C_{\text{dl}}$	
D:	$h_{\text{simp}}(s) = R_a^a / (1 + R_a^a C_{\text{simp}} s)$, $C_{\text{simp}} = C_{\text{dl}}$	

▲

Problem 7. Electric time constants.

Based on the two plots in Fig. 4, what are approximate values for time constants in the two plots (see u_a in the Wang-HOR plot and u_a in the Wang-ORR plot)?

	Problem 7: Approximate electric time constants for the anode and the cathode, which pair is <i>reasonable</i> ?	Choice
A:	$T_a \approx 0.5 \text{ ms}$, $T_c \approx 0.1 \text{ s}$	
B:	$T_a \approx 2 \text{ ms}$, $T_c \approx 10 \text{ s}$	
C:	$T_a \approx 0.01 \text{ s}$, $T_c \approx 0.5 \text{ s}$	
D:	$T_a \approx 100 \text{ ms}$, $T_c \approx 2 \text{ s}$	

▲

Problem 8. Anode species balance.

Figure 5 illustrates the most common gas subsystem for a fuel cell: hydrogen is supplied from a high pressure storage system via an inlet anode pressure controller, alternatively a flow controller. Effluent hydrogen from the anode volume is returned to the inlet via a fan (low anode pressure) or compressor (high anode pressure). The purge valve is normally closed, but can be opened for a short time to purge out accumulated impurities in the hydrogen gas.

Oxygen is normally supplied via atmospheric air (approximately 20% oxygen, 80% nitrogen), either by fan (low cathode pressure) or compressor (high cathode pressure). The fan/compressor flow rate can be controlled (speed control), and the effluent volume is controlled to pressure p_c^e (so-called back-pressure controller). If pure oxygen were supplied from a high pressure storage tank (conceptually possible), the oxygen feed system would have to be similar to the hydrogen supply system in the figure.

Assume that the well mixed anode volume V_a contains only $\text{H}_2\text{O} + \text{H}_2$, and that the water content is fixed to be $p_{\text{H}_2\text{O}}^a = \frac{1}{2}p_{\text{H}_2\text{O}}^{\text{sat}}(T)$ where the saturation pressure is given by Antoine's model as in Problem. 3.

Assuming ideal gas and constant temperature, and known influent $\dot{n}_{\text{H}_2}^i$, the hydrogen species balance leads to the following model for hydrogen pressure:

	Problem 8: Dynamic model for hydrogen pressure under given conditions — which model is <i>correct</i> ?	Choice
A:	$\frac{dp_{\text{H}_2}}{dt} = \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^e - \frac{RT}{V_a} \dot{n}_{\text{H}_2}^i - \frac{RT}{V_a} \dot{n}_{\text{H}_2}^g$	
B:	$\frac{dp_{\text{H}_2}}{dt} = \frac{RT}{V_a} \dot{n}_{\text{H}_2}^i - \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^e + \frac{RT}{V_a} \dot{n}_{\text{H}_2}^g$	
C:	$\frac{dp_{\text{H}_2}}{dt} = \frac{RT}{V_a} \dot{n}_{\text{H}_2}^i - \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^e$	
D:	$\frac{dp_{\text{H}_2}}{dt} = \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^i + \frac{V_a}{RT} \dot{n}_{\text{H}_2}^i - \frac{RT}{V_a} \dot{n}_{\text{H}_2}^g$	

▲

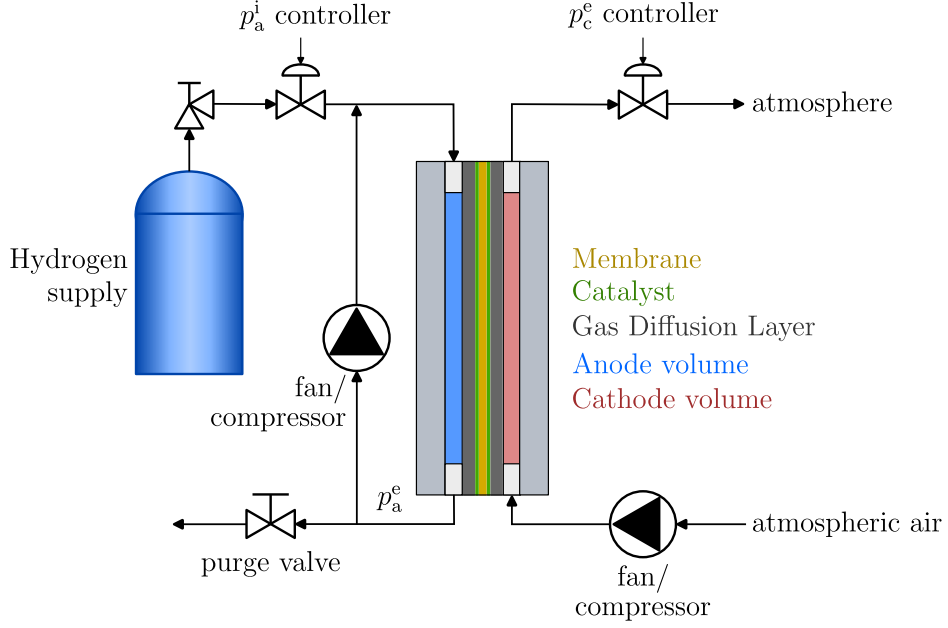


Figure 5: Gas subsystem for membrane.

Problem 9. Anode effluent flow rate.

Hydrogen is a precious fuel, and should not be wasted. For the gas subsystem in Fig. 5, assume for simplicity that $\dot{n}_{\text{H}_2}^i$ is a known/controlled function of time. Assume furthermore that the inlet pressure p_a^i to the anode volume is known and constant, and that a fan or compressor manages to keep a constant pressure increase in the *recycle* flow so that $\Delta p_a^{\text{rc}} = p_a^i - p_a^e$ is constant. From Bernoulli's steady state energy balance, a typical valve model for the effluent flow rate would be

$$\dot{V}_a^e = K_a \cdot \sqrt{\frac{\Delta p_a^e}{p^o}}. \quad (19)$$

Consider the term $\dot{V}_a^e$ in Problem 8. Modify the model in Eq. 19 so that (i) the flow unidirectional (i.e., only can flow out), and (ii) give expressions for Δp_a^e .

	Problem 9: Modified anode effluent flow rate $\dot{V}_a^e$ — which set of expressions is <i>correct</i> ?	Choice
A:	$\dot{V}_a^e = K_a \cdot \sqrt{\max(\Delta p_a^e, 0) / p^o}$; $\Delta p_a^e = p_a - p_a^i$; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$	
B:	$\dot{V}_a^e = K_a \cdot \sqrt{\min(\Delta p_a^e, 0) / p^o}$; $\Delta p_a^e = p_a - p_a^i$; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$	
C:	$\dot{V}_a^e = K_a \cdot \sqrt{\max(\Delta p_a^e, 0) / p^o}$; $\Delta p_a^e = p_a - p_a^e$; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$; $p_a^e = p_a^i - \Delta p_a^{\text{rc}}$	
D:	$\dot{V}_a^e = K_a \cdot \sqrt{\min(\Delta p_a^e, 0) / p^o}$; $\Delta p_a^e = p_a - p_a^e$; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$; $p_a^e = p_a^i - \Delta p_a^{\text{rc}}$	

▲

Problem 10. Anode hydrogen generation.

Consider the term $\dot{n}_{\text{H}_2}^g$ in Problem 8. With $\nu = (\nu_{\text{H}_2}, \nu_{\text{O}_2}, \nu_{\text{H}_2\text{O}})$, what is the correct expression for $\dot{n}_{\text{H}_2}^g$ for a single fuel cell?

	Problem 10: Which of the following expressions for hydrogen generation $\dot{n}_{\text{H}_2}^g$ is <i>correct</i> ?	Choice
A:	$\dot{n}_{\text{H}_2}^g = \nu_{\text{H}_2} \cdot i / F$	
B:	$\dot{n}_{\text{H}_2}^g = -\nu_{\text{H}_2} \cdot i / F$	
C:	$\dot{n}_{\text{H}_2}^g = \nu_{\text{H}_2} \cdot i / (F \nu_{e-})$	
D:	$\dot{n}_{\text{H}_2}^g = \nu_{\text{H}_2} \cdot i / F$	

▲

Problem 11. Anode diffusion.

There is a possible gas pressure drop from the anode volume to the surface of the catalyst/membrane due to diffusion resistance through the Gas Diffusion Layer (Fig. 5), etc. With only water vapor and hydrogen in the anode volume, a possible expression for the resulting pressure drop in hydrogen is

$$\frac{p_{\text{H}_2}^{\text{as}}}{p_{\text{H}_2}^{\text{a}}} = \frac{p_{\text{a}} - p_{\text{H}_2\text{O}}^{\text{a}} \cdot \exp(f_{\text{a}}(T) \cdot i'')}{p_{\text{a}} - p_{\text{H}_2\text{O}}^{\text{a}}} \quad (20)$$

where superscripts a and as refer to *bulk* anode volume and *anode surface* (i.e., at the catalyst/membrane surface), $p_{\text{H}_2\text{O}}^{\text{a}} = \frac{1}{2} p_{\text{H}_2\text{O}}^{\text{sat}}(T)$, $f_{\text{a}}(T) > 0$ is a known function of temperature (involving diffusion coefficients, etc.), while i'' is the current density in the membrane.

What modifications to the model is required if this diffusion term in Eq. 20 is important?

	Problem 11: Which of the following statements about the effect of anode diffusion is <i>correct</i> ?	Choice
A:	Diffusion drop in pressure has no effect on the model: a reduction in hydrogen pressure results in an increase in the water pressure, and these changes cancel out.	
B:	The anode volume model is as found above in Problems 8–10. However, the pressure $p_{\text{H}_2}^{\text{as}}$ should be used in the electric model to compute the fuel cell voltage.	
C:	The partial pressure of hydrogen in Problem 8 must be changed to $p_{\text{H}_2}^{\text{as}}$.	
D:	It is not necessary to modify the model because the controller for hydrogen feed will cancel out the effect.	

▲

Problem 12. Cathode oxygen balance.

Assume that the well mixed cathode volume V_{c} contains $\text{H}_2\text{O} + \text{O}_2 + \text{N}_2$, and that the water content is fixed to be $p_{\text{H}_2\text{O}}^{\text{c}} = p_{\text{H}_2\text{O}}^{\text{sat}}(T)$ where the saturation pressure is given by Antoine’s model as in Problem. 3.

Assuming ideal gas and constant temperature, and known influent flow rate $\dot{V}_{\text{c}}^{\text{i}}$ of atmospheric air at pressure p_{atm} (containing oxygen and nitrogen; mole fraction $x_{\text{O}_2}^{\text{air}} \approx 0.2$) controlled by the fan/compressor, the oxygen species balance leads to the following model for oxygen pressure:

	Problem 12: Dynamic model for oxygen pressure under given conditions — which model is <i>correct</i> ?	Choice
A:	$\frac{dp_{O_2}}{dt} = \frac{x_{O_2}^{air} p_{atm}}{V_c} \dot{V}_c^i - \frac{p_{O_2}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} \dot{n}_{O_2}^g$	
B:	$\frac{dp_{O_2}}{dt} = \frac{p_{O_2}}{V_c} \dot{V}_c^i - \frac{x_{O_2}^{air} p_{atm}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} \dot{n}_{O_2}^g$	
C:	$\frac{dp_{O_2}}{dt} = \frac{x_{O_2}^{air} p_{atm}}{V_c} \dot{V}_c^i - \frac{p_{O_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} \dot{n}_{O_2}^g$	
D:	$\frac{dp_{O_2}}{dt} = \frac{x_{O_2}^{air} p_{atm}}{V_c} \dot{V}_c^i - \frac{p_{O_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} (\dot{n}_{H_2O} - \dot{n}_{N_2})$	

▲

Problem 13. Cathode nitrogen balance.

With the assumptions in Problem 12, the nitrogen species balance leads to the following model for nitrogen pressure:

	Problem 13: Dynamic model for nitrogen pressure under given conditions — which model is simplest and <i>correct</i> ?	Choice
A:	$\frac{dp_{N_2}}{dt} = \frac{x_{O_2}^{air} p_{atm}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e$	
B:	$\frac{dp_{N_2}}{dt} = \frac{x_{O_2}^{air} p_{atm}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} (\dot{n}_{H_2O}^g - \dot{n}_{O_2}^g)$	
C:	$\frac{dp_{N_2}}{dt} = \frac{(1-x_{O_2}^{air}) p_{atm}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e$	
D:	$\frac{dp_{N_2}}{dt} = \frac{(1-x_{O_2}^{air}) p_{atm}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} \dot{n}_{N_2}^g$	

▲

Problem 14. Cathode effluent flow rate.

We will assume that $\dot{V}_c^i$ is a known function of time. From Bernoulli's steady state energy balance, a typical valve model for the *effluent* flow rate would be

$$\dot{V}_c^e = K_c \cdot \sqrt{\frac{\Delta p_c^e}{p^o}}. \quad (21)$$

Consider the term $\dot{V}_c^e$ in Problems 12–13. Modify the model in Eq. 21 so that (i) the flow unidirectional (i.e., only can flow out), and (ii) give expressions for Δp_c^e .

	Problem 14: Modified cathode effluent flow rate $\dot{V}_c^a$ — which set of expressions is <i>correct</i> ?	Choice
A:	$\dot{V}_c^e = K_c \cdot \sqrt{\max(\Delta p_c^e, 0) / p^o}; \Delta p_c^e = p_c - p_{atm}; p_c = p_{O_2} + p_{N_2} + p_{H_2O}^c$	
B:	$\dot{V}_c^e = K_c \cdot \sqrt{\min(\Delta p_c^e, 0) / p^o}; \Delta p_c^e = p_c - p_{atm}; p_c = p_{O_2} + p_{N_2} + p_{H_2O}^c$	
C:	$\dot{V}_c^e = K_c \cdot \sqrt{\max(\Delta p_c^e, 0) / p^o}; \Delta p_c^e = p_c - p_{atm}; p_c = p_{O_2} + p_{N_2}$	
D:	$\dot{V}_c^e = K_c \cdot \sqrt{\min(\Delta p_c^e, 0) / p^o}; \Delta p_c^e = p_c - p_{atm}; p_c = p_{O_2} + p_{N_2}$	

▲

Problem 15. Cathode generation.

Consider the term $\dot{n}_{O_2}^g$ in Problem 12. With $\nu = (\nu_{H_2}, \nu_{O_2}, \nu_{H_2O})$, what is the correct expression for $\dot{n}_{O_2}^g$ for a single fuel cell?.

	Problem 15: Which of the following expressions for oxygen generation $\dot{n}_{O_2}^g$ is <i>correct</i> ?	Choice
A:	$\dot{n}_{O_2}^g = \nu_{O_2} \cdot i / (F \nu_{e-})$	
B:	$\dot{n}_{O_2}^g = -\nu_{O_2} \cdot i / (F \nu_{e-})$	
C:	$\dot{n}_{O_2}^g = \nu_{O_2} \cdot i / F$	
D:	$\dot{n}_{O_2}^g = -\nu_{O_2} \cdot i / F$	



Problem 16. Cathode diffusion/current limitation.

There is a possible gas pressure drop from the cathode volume to the surface of the catalyst/membrane due to diffusion resistance through the Gas Diffusion Layer (Fig. 5), etc. With water vapor, oxygen, and nitrogen in the cathode volume, a possible expression for the resulting pressure drop in oxygen pressure is

$$\frac{p_{\text{O}_2}^{\text{cs}}}{p_{\text{O}_2}^{\text{c}}} = \frac{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}} \exp(f_{\text{c}}(T) \cdot i'')}{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}}} \quad (22)$$

where superscripts c and cs refer to *bulk* cathode volume and *surface* (i.e., at the catalyst/membrane surface), $p_{\text{H}_2\text{O}}^{\text{c}} = p_{\text{H}_2\text{O}}^{\text{sat}}(T)$, $f_{\text{c}}(T) > 0$ is a known function of temperature (involving diffusion coefficients, etc.), while i'' is the current density in the membrane.

In the literature, it is common to write this reduction in partial pressure as

$$\frac{p_{\text{O}_2}^{\text{cs}}}{p_{\text{O}_2}^{\text{c}}} \approx 1 - \frac{i''}{i_{\ell}''}$$

where i_{ℓ}'' is a *limiting* current density.

Find an expression for i_{ℓ}'' based on (linearization of) Eq. 22.

	Problem 16: Which of the following expressions for i_{ℓ}'' is <i>consistent</i> with Eq. 22?	Choice
A:	$i_{\ell}'' = \frac{p_{\text{N}_2}^{\text{c}}}{p_{\text{O}_2}^{\text{c}}} \cdot f_{\text{c}}(T)$	
B:	$i_{\ell}'' = \frac{p_{\text{N}_2}^{\text{c}}}{p_{\text{O}_2}^{\text{c}}} \cdot \frac{1}{f_{\text{c}}(T)}$	
C:	$i_{\ell}'' = \frac{p_{\text{O}_2}^{\text{c}}}{p_{\text{N}_2}^{\text{c}}} \cdot f_{\text{c}}(T)$	
D:	$i_{\ell}'' = \frac{p_{\text{O}_2}^{\text{c}}}{p_{\text{N}_2}^{\text{c}}} \cdot \frac{1}{f_{\text{c}}(T)}$	



Problem 17. Maximal Fuel Cell efficiency.

Fuel Cell efficiency η is typically found as the produced electric power divided by the enthalpy of reaction,

$$\eta = \frac{u \cdot i}{\Delta_{\text{r}} \tilde{H} \cdot \dot{n}_{\text{H}_2\text{O}}^{\text{g}}}.$$

Here, current i is given as

$$i = -F \cdot \dot{n}_{\text{e}^-}$$

where $\dot{n}_{\text{e}^-}$ is the molar flow rate of electrons, and the generation of product $\dot{n}_{\text{H}_2\text{O}}^{\text{g}}$ is related to $\dot{n}_{\text{e}^-}$ as

$$\dot{n}_{\text{H}_2\text{O}}^{\text{g}} = \frac{\dot{n}_{\text{e}^-}}{\nu_{\text{e}^-}}.$$

In the limit $i \rightarrow 0^+$, $u = u_{\text{n}}$ where Nernst voltage u_{n} is as given in Problem 2. Denote the resulting maximal efficiency by η_0 [zero current],

$$\eta_0 = \frac{u_{\text{n}} \cdot i}{\Delta_{\text{r}} \tilde{H} \cdot \dot{n}_{\text{H}_2\text{O}}^{\text{g}}}.$$

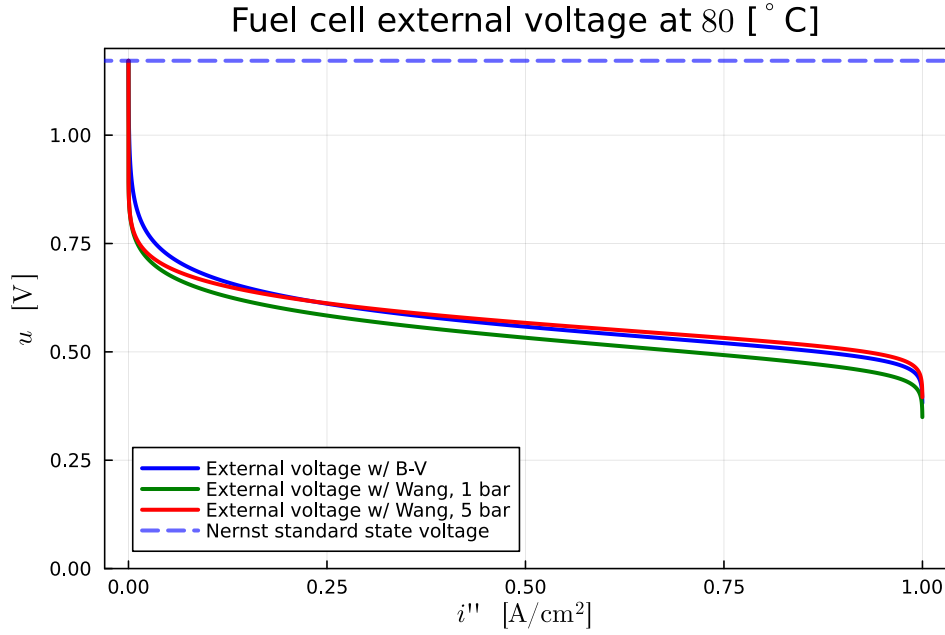


Figure 6: Typical external voltage of fuel cell: (i) using Butler-Volmer activation voltage, (ii)-(iii) using a more realistic activation voltage due to Wang, at 1 and 5 bar, respectively, (iv) with zero current/Nernst voltage (blue, dashed) .

For simplicity, we consider a reference temperature of 25 °C, which gives

$$\Delta_r \tilde{G} = \Delta_r \tilde{G}^\circ = 1 \cdot \Delta_f \tilde{G}_{\text{H}_2\text{O}} = -228.572 \text{ kJ/mol}$$

while

$$\Delta_r \tilde{H}^\circ = -241.785 \text{ kJ/mol}.$$

Based upon the above considerations, what is the theoretical maximal efficiency of a fuel cell at standard state (i.e., pressures $p_j = p^\circ$) and with (extremely) low current throughput i , when we assume that both reactants and product are in gas phase?

	Problem 17: Approximate fuel cell efficiency under the above considerations — which maximal efficiency η_0 is <i>correct</i> ?	Choice
A:	$\eta_0 \approx 0.4$	
B:	$\eta_0 \approx 0.95$	
C:	$\eta_0 \approx 0.83$	
D:	$\eta_0 \approx 0.28$	

▲

Problem 18. Practical Fuel Cell efficiency.

Figure 6 shows how the external voltage u in the efficiency expression of Problem 17 varies with current density i'' (and thus with i).

Based on the above, how much is the practical efficiency $\eta_{i''}$ of a fuel cell *reduced* compared to the maximal efficiency, say at $i'' = 0.5 \text{ A/cm}^2$?

	Problem 18: Approximate fuel cell efficiency under the above considerations — which efficiency $\eta_{0.5}$ is <i>correct</i> ?	Choice
A:	$\eta_{0.5} \approx \eta_0$	
B:	$\eta_{0.5} \approx 0.49 \cdot \eta_0$	
C:	$\eta_{0.5} \approx 0.61 \cdot \eta_0$	
D:	$\eta_{0.5} \approx 0.83 \cdot \eta_0$	

▲

Problem 19. Consequence of efficiency loss.

In Problems 17–18, we have seen that a fuel cell does not have 100% conversion of reactants into electric power.

What is the consequence of imperfect efficiency η ?

	Problem 19: Which of the following statements is <i>correct</i> ?	Choice
A:	Both with $i'' \rightarrow 0^+$ (η_0) and for finite i'' ($\eta_{i''}$) will the efficiency loss lead to heating of the fuel cell. It is necessary to actively control the cooling of the fuel cell to (i) avoid overheating, and (ii) manage the humidity of the membrane.	
B:	The efficiency loss with $i'' \rightarrow 0^+$ (η_0) is an <i>inherent</i> loss, that has no effect on the temperature of the system; there can be no loss when there is no current.	
C:	The efficiency loss for finite $i'' > 0$ ($\eta_{i''}$) leads to heating of the fuel cell, and some cooling is necessary. However, this does not influence the humidity of the system.	
D:	The efficiency loss for finite $i'' > 0$ ($\eta_{i''}$) leads to heating of the fuel cell, but the heating is automatically compensated for by the humidity/water balance.	

▲

Problem 20. Choice of operating conditions.

As seen from the expression for pressure correction to the standard state equilibrium voltage, increasing the pressure (if possible) should give a higher equilibrium voltage, and hence more power out of the fuel cell.

The Butler-Volmer expression for activation voltage used in the group project was pressure independent; however, a more accurate (and complex) expression for the cathode activation, Fig. 7, shows that the voltage loss u_a for a *given* current density i'' *decreases* with pressure.

In summary: the equilibrium voltage *increases* with higher electrode pressures, and the activation *losses* are *reduced* when current starts to flow. This means that there is a *double advantage* in operating at higher pressure.

Which of the following arguments for choice of operating pressures has potential safety issues?

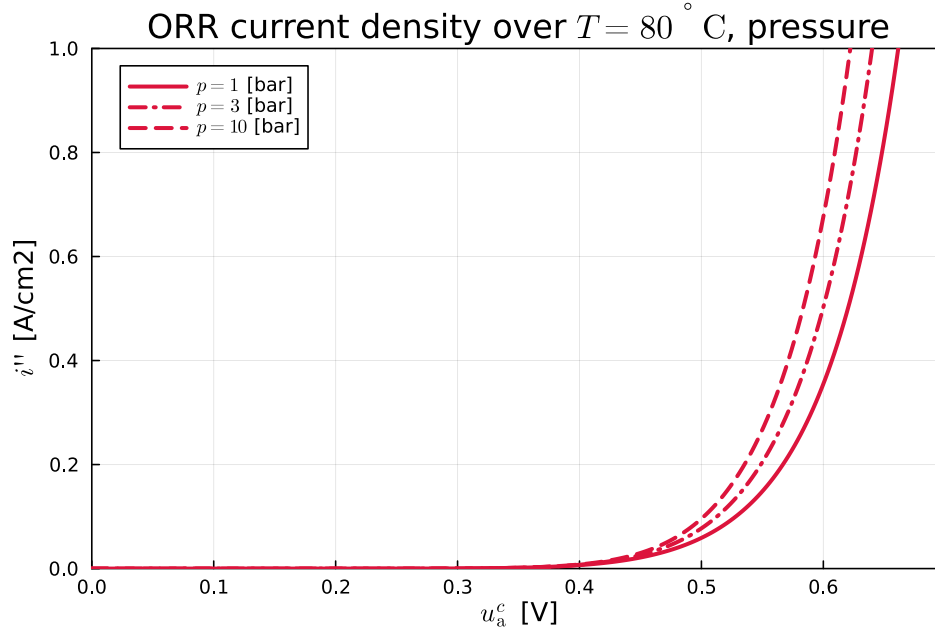


Figure 7: Pressure dependence of activation voltage loss relation.

	Problem 20: Choice of operating pressures for fuel cell — which argument has <i>safety issues</i> ?	Choice
A:	Operating at a high pressure requires power to compress the reactants (hydrogen, oxygen). The chosen pressure conditions must balance the increased electric power produced vs. the increased power needed to compress the reactants.	
B:	Increased pressures in the electrode volumes must be balanced against the possibility of gas permeation through the membrane; higher pressure may require thicker membrane.	
C:	Operating at a high pressure may require more expensive equipment (thicker membrane to avoid gas permeation, also leading to higher resistive voltage loss u_r), more solid build, etc., and the increased design/purchase cost must be balanced against increased power production.	
D:	The best way to increase the power is to increase the Nernst voltage by feeding oxygen to the cathode instead of air. This way, the partial pressure of oxygen can be made much higher while still having the same electrode pressures, $p_a = p_c$.	

▲

— END OF EXAM —